Predicting Customer Purchase Behavior Using Predictive Modeling Techniques

MIS 5560

Group 5

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Project Description

This project focuses on predicting customer purchase behavior for an online retail store using transactional and behavioral data from a Kaggle Dataset titled "eCommerce behavior from multi-category store." The dataset contains "user sessions" collected from seven months, from October 2019 to April 2020. Each "user session" represents a relation between product and user, such as visits, product views, duration of session, category of product, etc.

The primary business problem that will be addressed in this project is the following, "online retailers want to increase the conversion rate of visitors to paying customers."

The goal of this project is to use predictive models to predict the likelihood of purchase, understand key drivers of purchase behavior, and compare model performance to determine the most efficient approach for targeting marketing efforts that will assist online retailers. We will use the following models: Linear regression, logistic regression, decision tree, lasso-regression, random forest, and neural net.

Data Exploration and Preprocessing

During data preprocessing, we made several adjustments to improve data quality, reduce redundancy, and enhance model performance. First, we removed variables that did not provide meaningful variation. Specifically, "event_date", "event_dayofweek", and "is_weekend" were dropped due to all observations being on the same day, October 1st, making these variables constant and uninformative for modeling. Additionally, we also removed "remove_from_cart_count" as it contained only zero values across all observations.

To improve computational efficiency, given the large dataset size of millions of rows, we initially worked with a smaller sample dataset. The smaller data set was achieved by taking a random selection of 257, 813 observations. This allowed for faster exploration, visualization, and transformation before applying the same preprocessing steps to the full October dataset.

We also adjusted new features to better capture user behavior. The hour-based variable was transformed into binary indicator variables, such as “is_afternoon” and “is_night”, with true or false values, to align with modeling approaches like logistic regression and improve interpretability.

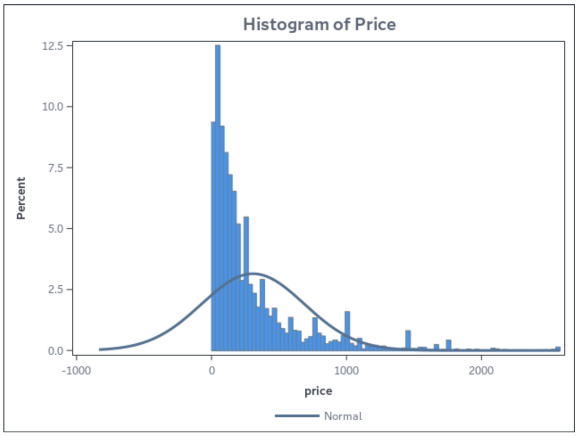
Finally, after cleaning and transforming the data, the dataset was prepared for modeling by ensuring all variables were in appropriate formats and ready for techniques such as logistic regression, linear regression, decision trees, and other machine learning models.

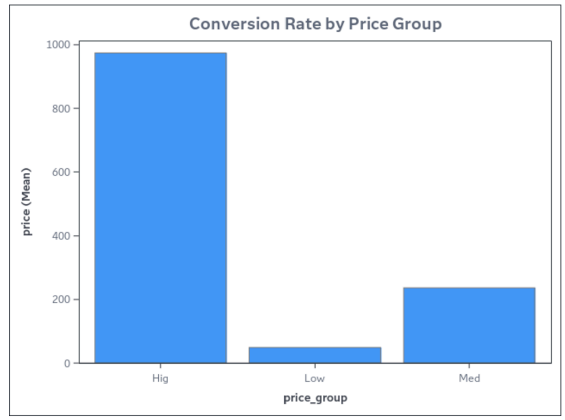
The target variable in our analysis is “purchase,” the binary variable created from the “event_type” column. We used the following to classify whether a purchase occurred or not (1=purchase, 0=not a purchase).

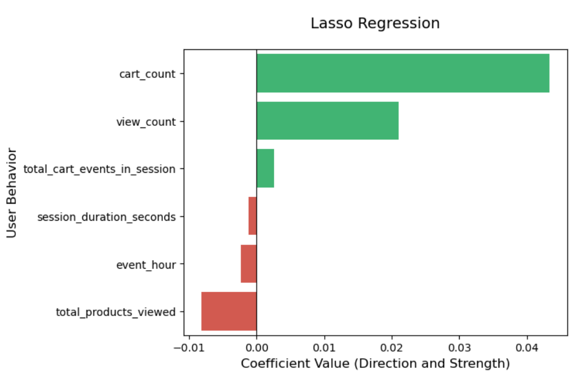
The primary independent variables we used in the linear regression model and the decision tree model are product_id and price. In our more advanced models, we used additional independent variables: view_count, cart_count, remove_from_cart_count, Total_events_in_session, total_products_viewed, Unique_categories_viewed, total_cart_events_in_session, Price, session_duration_seconds,event_hour, event_dayofweek, and is_weekend.

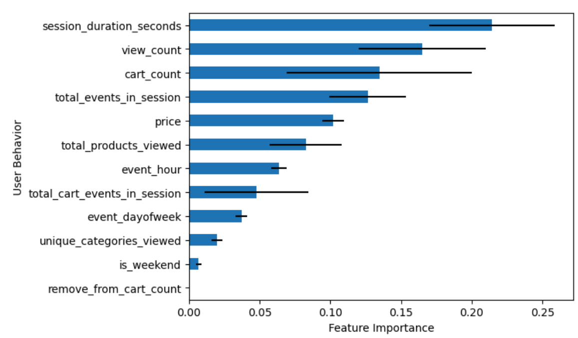
Several graphs/plots were used to explore the data. First, we created a histogram of price to examine the distribution of product prices and identify skewness and outliers. Next, we created a bar chart that shows the conversion rate by price group, by creating three price groups (high, low, and medium), to determine where the majority of the price fell. Lastly, we did Lasso

Regression and Random Forest plots that illustrated the coefficient value of various independent variables, as well as the feature importance.









Proc Means was used to check for any missing values, and the output signified that there were no significant missing values; therefore, nothing was needed to address them. Also, as we mentioned above, the following variables were deemed redundant and were removed from the dataset: event_date, event_dayofweek, and is_weekend. Overall, we did not have any major data errors/missing data that required correction.

In our linear regression model, we checked for multicollinearity, or redundancy, through VIF, Variance Inflation Factor. Our report indicated a value of 1.06, meaning that there was no significant correlation between variables and no redundant variables. This meant there was no strong multicollinearity detected which would require further problem-solving.

The dataset was split into training and validation sets, with 60% of the data in the training set and 40% of the data in the validation set. We used the training set to build the model and evaluated its performance using the validation set.

Model

Linear regression was used as a baseline model due to its simplicity and ease of interpretation. The following shows that as price increase, the probability of purchase slightly decreases. Purchase = 0.03690 - $8.48 * 10^{-10}$ * (product_id) - $5.53 * 10^{-6}$ * (price)

Least Absolute Shrinkage and Selection Operator, better known as Lasso, is similar to linear regression, but penalizes variables based on the size of their coefficients. The model shrinks the coefficients of less important variables to zero, thus leaving the more influential variables. In short, the model received all numeric variables, ran a regression, and output the ones that contributed the most. Logistic regression and neural networks are more appropriate for classification problems and can capture more complex relationships. A decision tree model was used because it is well-suited for binary classification problems and can capture nonlinear relationships while remaining easy to interpret. It does not have a single mathematical equation like linear regression. Instead, it works by splitting the data into branches based on decision rules derived from the independent variables. In this case, the model creates splits based on values of product_id and price to classify whether a purchase will occur (1) or not (0). Each split is chosen to maximize the separation between the two classes. Lastly, a random forest model improved predictive accuracy and reduced variance by aggregating multiple trees.

Model Specifications

Model	Sample Size	Dependent	Independent	Metrics
Linear Regression	**257,813**	**purchase**	**product_id, price**	**R^2, t-values, Adjusted R^2**
Logistic Regression	**1000000**	**purchase**	**view_count, cart_count, remove_from_cart_count, Total_events_in_session, total_products_viewed, Unique_categories_viewed, total_cart_events_in_session, Price, session_duration_seconds,event_hour, event_dayofweek, is_weekend**	**Confusion matrix, coefficients, ROC**
Decision Tree	**103,126**	**purchase**	**product_id, price**	**Confusion matrix, accuracy score**
Lasso Regression	**257,813**	**purchase**	**view_count, cart_count, remove_from_cart_count, Total_events_in_session, total_products_viewed, Unique_categories_viewed, total_cart_events_in_session, Price, session_duration_seconds,event_hour, event_dayofweek, is_weekend**	**RMSE & Coefficients**
Random Forest	**257,813**	**purchase**	**view_count, cart_count, remove_from_cart_count,**	**Importance of Variables**

			Total_events_in_session, total_products_viewed, Unique_categories_viewed, total_cart_events_in_session, Price, session_duration_seconds,event_hour, event_dayofweek, is_weekend	
Neural Net	**1000000**	**purchase**	**view_count, cart_count, remove_from_cart_count, Total_events_in_session, total_products_viewed, Unique_categories_viewed, total_cart_events_in_session, Price, session_duration_seconds,event_hour, event_dayofweek, is_weekend**	**Confusion Matrix, Sample Predictor**

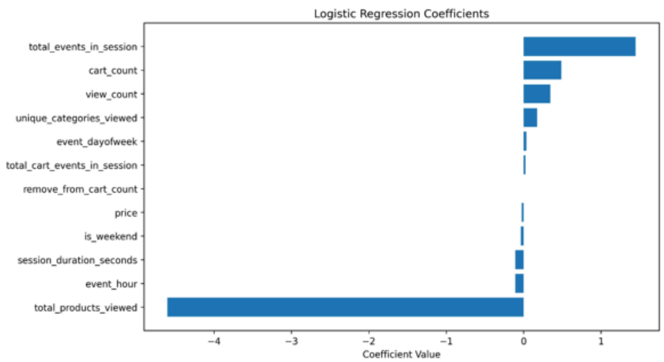
Model Comparison

Linear Regression

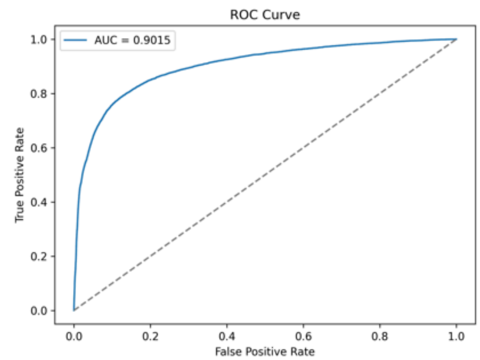
- F-value = 551.56
- P-value < 0.0001
- This indicates that the model is statistically significant
- $R^2 = 0.0043$
- This indicates that the model can only explain a very small portion of the variation in purchase behavior.

Logistic Regression

The logistic regression model provides a clear picture of e-commerce behavior. The model results in a high accuracy(97.5%) and strong ROC AUC(.9). It shows classic e-commerce behavior where long sessions with a lot of products_viewed (endless browsing) are fare less likely to purchase with a co-efficient of -4.6 while event_count (adding to cart, viewing, etc) has the opposite effect with a co-efficient of 1.44. Smaller but noticeable effects are identified in price, session_duration and time/day of week. The confusion matrix shows high reliability identifying non purchasers while less accurate and more conservative identifying purchasers.



	Pred 0	Pred 1
Actual 0	389020	1055
Actual 1	8726	1199



Metric	Value
Accuracy	0.975548
ROC AUC	0.901513

<u>Decision Tree</u>

The decision tree model achieved an accuracy of approximately 97.4%. However, despite the high accuracy, the confusion matrix revealed that the model failed to correctly classify any purchase cases and instead predicted all observations as non-purchases. This indicates that the model is heavily influenced by class imbalance and does not effectively identify customers who are likely to make a purchase.

Overall, while the decision tree provides a more flexible and interpretable structure than linear regression, it still does not fully address the business problem. An ensemble model such as a random forest would likely improve performance by combining multiple decision trees to reduce bias and better capture patterns in the data. Therefore, more advanced models are recommended to improve prediction accuracy and better support targeted marketing strategies.

Lasso Regression

- RMSE = .1474
- Coefficients:
  - view_count: 0.0210
  - cart_count: 0.0434
  - total_products_viewed: -0.0082
  - total_cart_events_in_session: 0.0026
  - session_duration_seconds: -0.0012
  - event_hour: -0.0023

Random Forest

	feature	importance	std
2	remove_from_cart_count	0.000000	0.000000
11	is_weekend	0.006798	0.001687
5	unique_categories_viewed	0.019619	0.003886
10	event_dayofweek	0.037087	0.003972
6	total_cart_events_in_session	0.047575	0.036785
9	event_hour	0.063642	0.005650
4	total_products_viewed	0.082645	0.025428
7	price	0.101999	0.007754
3	total_events_in_session	0.126405	0.026972
1	cart_count	0.134654	0.065489
0	view_count	0.165154	0.044865
8	session_duration_seconds	0.214422	0.044494

Linear Regression

The REG Procedure
Model: model1
Dependent Variable: purchase

Number of Observations Read	257813
Number of Observations Used	257813

Analysis of Variance

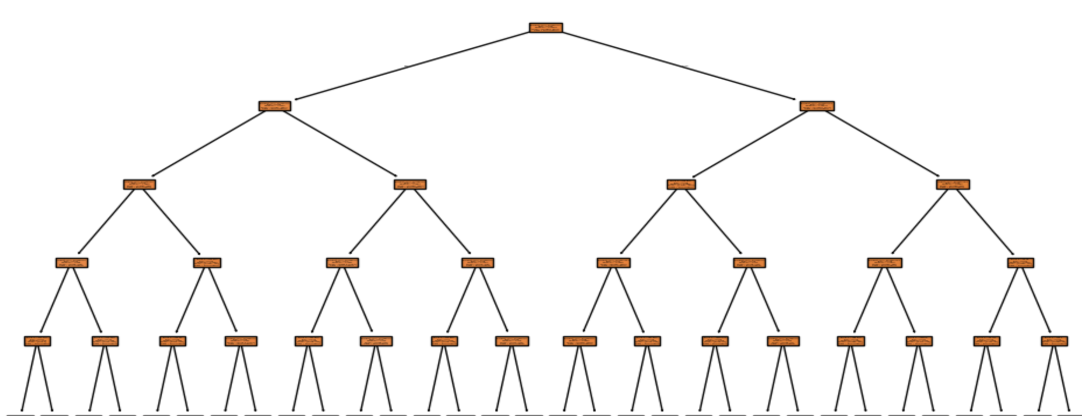
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	27.21403	13.60702	551.56	<.0001
Error	257810	6360.17329	0.02467		
Corrected Total	257812	6387.38732			

Root MSE	0.15707	R-Square	0.0043
Dependent Mean	0.02542	Adj R-Sq	0.0043
Coeff Var	617.84988		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Standardized Estimate	Variance Inflation
Intercept	1	0.03690	0.00053620	68.82	<.0001	0	0
product_id	1	-8.4848E-10	2.55847E-11	-33.16	<.0001	-0.06723	1.06407
price	1	-0.00000535	8.395232E-7	-6.37	<.0001	-0.01292	1.06407

Decision tree



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Confusion Matrix:
 [[100443      0]
 [  2683      0]]
Accuracy: 0.9739832825863507

Feature Importance:
price : 0.147838175364426
product_id : 0.8521618246355739
```

<u>Neural Net</u>

The NN model performed well with the overall accuracy among all three around 98%. The small and medium NN performed the best at 98.62 and 98.63, but the large takes a dive in accuracy. This shows that once the model is detailed enough to capture the decision boundary, adding more layers or neurons only increases training time and risk of overfitting without boosting predictive power by inserting unnecessary noise. The medium is more accurate, but the small was able to predict more correct purchases. However, both perform well and show significant improvements from other models.

Model	Accuracy	Elapsed (sec)
Small NN	0.98619	88.97384130000137
Medium NN	0.986295	194.3005497000413
Large NN	0.985345	499.8412876999937

Small	Predicted 0	Predicted 1
Actual 0	387982	1960
Actual 1	3564	6494

Medium	Predicted 0	Predicted 1
Actual 0	388424	1518
Actual 1	3964	6094

Large	Predicted 0	Predicted 1
Actual 0	388099	1843
Actual 1	4019	6039

Summary

Throughout this project, several lessons were learned. First, that preprocessing of data and the quality of data you're working with, are critical to the overall outcome of your analysis. It was necessary for us to remove irrelevant or constant variables, such as columns with all zeroes, to simplify the data we were working with and make it easier to assess. Similarly, choosing a random selection of the dataset, to shrink the size, made it more efficient to work with.

Secondly, we learned how important it is to choose the right independent variables. As we found throughout our analysis, some variables did not have the desired outcome we were looking for, or did not have an outcome at all. We were able to identify important variables to use through correlation checks and lasso regression. Using the correct independent variables provided us with improved interpretability, allowing us to analyze the data in relation to our business problem.

Lastly, we learned that different models have different strengths and weaknesses. For example, the logistic regression performed well for purchase vs no purchase due to it having strong interpretability and binary outcomes. On the other hand, linear regression was limited for classification, but was useful for continuous prediction. Neural network captured complex patterns, but the model required more tuning and it had a weaker interpretability. Overall, it was proven that while the more complex models performed better, there was not a single model that was best on its own. In order to conduct a more full analysis, it was more beneficial to utilize several models.

Resources

Kechinov, Michael. *E-commerce Behavior Data from Multi-Category Store*. Kaggle, https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store. Accessed 11 Apr. 2026.

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